Chapter 1: Historical Background and Overview

Glycobiology is the study of the structure, biosynthesis, biology, and evolution of glycans plus the proteins that recognize them. The central argument is that the DNA→RNA→protein paradigm is incomplete: every cell and many macromolecules carry covalently attached glycans, and because glycans are among the major posttranslational modifications, they help explain how a modest gene count produces enormous biological complexity. Glycans are framed as the "dark matter" of biology; abundant, critical, under-incorporated into the standard model.

Key foundations introduced: the glycocalyx (the dense sugar coat on all cells), the SNFG symbol nomenclature, and the major glycoconjugate classes. The line most relevant to me: glycan structures are not encoded directly in the genome, they are secondary gene products built by competing glycosidases and glycosyltransferases, so even full knowledge of gene expression cannot predict the exact glycans a cell makes. This non-template nature is *why* experimental confirmation (the thing we're curating for) matters. It also introduces microheterogeneity: a single polypeptide from one gene can exist as many "glycoforms," with variable glycans at a given site (and sometimes no glycan at all).

Chapter 2: Monosaccharide Diversity

Monosaccharides are aldoses or ketoses; D/L configuration is set by the highest-numbered chiral carbon; ring closure creates the anomeric center (α vs β). The combinatorial point that drives the whole field: unlike amino acids and nucleotides, each monosaccharide can form α- or β-linkages to any of several hydroxyls on another sugar, and can branch, so three hexoses can theoretically yield thousands of trisaccharides versus six trimers for three amino acids.

Two curation-relevant distinctions live here. First, **glycation vs glycosylation**: glycation is a nonenzymatic Schiff-base reaction between reducing sugars and lysine residues, distinct from glycosylation, which forms a true glycosidic bond. If an abstract describes a glycation/Amadori product (common in diabetes literature), that is *not* an enzymatic glycosylation site. Second, the nine common vertebrate monosaccharides and their standard abbreviations worth knowing cold since they're the vocabulary our extracted sites might use?

Chapter 3: Oligosaccharides and Polysaccharides

Reinforces the N-glycan core and shows a typical O-glycan core, both as branched structures. N-glycans bond to the side-chain nitrogen of an Asn in the consensus NX(S/T); O-glycans bond to the oxygen of Ser or Thr, the two linkage chemistries we'll be distinguishing constantly. The chapter then tours structural/storage polysaccharides (cellulose vs starch as the α/β-linkage object lesson), GAGs, and the very diverse bacterial polysaccharides. <u>For our PTM-extension thinking, the recurring theme is that linkage and modification, not just composition, define function.</u>

Chapter 4: Cellular Organization of Glycosylation

Most secretory and membrane proteins move through the ER→Golgi→trans-Golgi-network pathway, where the major glycosylation reactions happen. The **topological rule** is the load-bearing idea: whatever faces the lumen of the ER/Golgi ends up facing the outside of the cell (or the inside of a lysosome/granule), and there are no well-documented exceptions. The corollary matters for distinguishing glycan classes: nuclear and cytoplasmic glycosylation is topologically reversed, those glycosyltransferases face the cytosol, which is why O-GlcNAc (Ch. 19) is mechanistically separate from the secretory N-/O-glycans. O-GlcNAc may be the single most common glycoconjugate in many cell types, a useful reminder when your pipeline encounters O-GlcNAc claims that shouldn't be lumped with secretory O-glycans.

Chapter 5: Glycosylation Precursors

Glycans are built from activated donors - nucleotide sugars (UDP-, GDP-, CMP-linked) and lipid-linked sugars like dolichol-P-mannose. Sialic acids are the only animal monosaccharides activated as CMP-mononucleotides, and iduronic acid has no nucleotide-sugar parent because it's made by epimerizing glucuronic acid after incorporation into GAG chains. Sources are dietary uptake, salvage from degraded glycans, and de novo synthesis; transporters (GLUT/SLC2A, SGLT/SLC5A) and lysosomal carriers move sugars around. Mostly background for curation, but it grounds why certain residue–donor pairings are or aren't chemically plausible.

Chapter 6: Glycosyltransferases and Glycan-Processing Enzymes

The classic frame is Roseman's **"one enzyme–one linkage"** hypothesis: most glycosyltransferases are highly specific for both donor and acceptor, with the B-blood-group α1-3 galactosyltransferase as the exemplar - it only acts on Gal already modified by α1-2 fucose. The chapter then catalogs the real-world exceptions (relaxed specificity, one enzyme/multiple linkages, bifunctional enzymes like the heparan-sulfate and hyaluronan synthases). <u>For our residue–PTM-compatibility-check feature idea (v2.0), this is the biochemical justification: enzyme specificity is what makes residue/linkage/sugar combinations predictable enough to flag implausible extractions.</u>

Chapter 7: Biological Functions of Glycans

Functions sort into three buckets: structural roles, energy metabolism, and information-carrying (recognition by glycan-binding proteins), with GBPs split into intrinsic (same organism) and extrinsic (pathogens, symbionts). Two honest caveats run through it: all the major theories of glycan function have supporting evidence and exceptions, and genetic glycosylation defects are often silent in cultured cells but catastrophic in whole organisms, i.e., function is mostly visible at the organism level.

Chapter 8: A Genomic View of Glycobiology

Introduces the glycome and the CAZy database, which classifies glycosyltransferases (GTs) and glycoside hydrolases (GHs) by sequence/fold rather than by reaction. Useful number: a few percent of any genome encodes GTs and GHs, and mammalian genomes encode an estimated 100–200 glycan-binding proteins. The practical caution for any sequence-based prediction (relevant to our ML pipeline): sequence similarity often predicts the anomeric linkage or broad glycan category, but not always the precise reaction.

Chapter 9: N-Glycans (core to our curation)

The sequon. The minimal acceptor sequence is Asn-X-Ser/Thr, where X is any amino acid except proline. All eukaryotic N-glycans begin with GlcNAcβ1-Asn. Critical curation nuance: ~70% of proteins contain the sequon and ~70% of sequons actually carry a glycan - so the sequon is necessary but not sufficient. Sequons encoded by mRNA are only "potential" N-glycan sites; proof that a glycan is actually present requires experimental evidence.

Edge cases that could (?) affect our validation logic: N-glycans occasionally occur at Asn-X-Cys, and rarely with a different third residue; an acidic X (Asp/Glu) reduces efficiency. So an extracted "non-canonical" N-site isn't automatically a false positive.

Shared core and three classes. Every eukaryotic N-glycan shares the core, and they fall into three types: oligomannose (only Man extends the core), complex (GlcNAc-initiated antennae), and hybrid (one arm Man, the other GlcNAc-initiated).

Biosynthesis (the en-bloc mechanism). Synthesis starts on dolichol-phosphate, builds a 14-sugar precursor, and transfers it en bloc to Asn during/after translocation into the ER lumen, then processes through ER and Golgi in a species-, cell-, protein-, and site-specific way. This is why the same site can show different glycoforms across cell types.

How sites get confirmed experimentally - directly relevant to what counts as evidence in an abstract: PNGase F releases most N-glycans and converts the Asn to Asp (PNGase A removes all). Endoglycosidase H releases oligomannose and hybrid but not complex N-glycans. The Asn→Asp deamidation signature (a +0.984 Da mass shift / N-to-D sequence change after PNGase F) is a classic mass-spec confirmation of an *occupied* site - good signal that an extracted site is experimentally validated rather than merely predicted. The Endo H sensitivity/resistance distinction is also frequently reported and tells you the glycan *class*, which may matter for your residue–PTM annotations.

Chapter 10: O-GalNAc Glycans (core to our curation)

Linkage and ambiguity. O-GalNAc glycans initiate with GalNAc α-linked to the hydroxyl of Ser or Thr; the sugars involved are GalNAc, Gal, GlcNAc, Fuc, and Sia, notably not Man, Glc, or Xyl. That exclusion list is a handy validation rule: a Man/Glc/Xyl-on-Ser/Thr claim is a *different* O-glycan class

(O-mannose, O-glucose, O-fucose - Ch. 13/18), not O-GalNAc. There's no sequon for O-GalNAc - site prediction is sequence-context-based, not motif-based, which makes experimental confirmation even more essential than for N-glycans.

Four core structures. A single GalNAc on Ser/Thr is the Tn antigen; adding β1-3Gal makes core 1 (the most common, = the T/Thomsen–Friedenreich antigen); core 2 adds β1-6GlcNAc to core 1; cores 3 and 4 are GlcNAc-based and largely restricted to GI and bronchial mucins. Tn, T, and their sialylated forms (sialyl-Tn, sialyl-T) are usually cryptic but elevated in cancer cells - so cancer-context abstracts often report these specifically.

Mucins and the PTS domain. Mucins carry the densest O-GalNAc, concentrated in PTS domains (rich in Pro, Thr, Ser; formerly "VNTR"), where glycans can be 50–80% of molecular weight, producing the extended "bottle-brush" conformation. Humans have ~20 mucin genes.

Experimental release/confirmation methods to recognize as evidence: β-elimination (mild alkali) releases O-glycans and is preferred when N-glycans are also present, since N-glycans resist mild conditions; O-glycanase releases unsubstituted core 1; sialidase + O-glycanase releases most simple core 1 glycans. Unlike N-glycans, there's no single enzyme that releases all intact complex O-GalNAc glycans, which is partly why O-site evidence in the literature is messier and more heterogeneous than N-site evidence.